

Plasma Proteomic Correlates of Mechanically Affected Lung and a Derived Risk Score for Emphysema Progression in COPDGene

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Abstract

Background. Mechanically Affected Lung (MAL) is a CT-derived continuous score quantifying the volume of lung parenchyma under destructive mechanical stress. Although MAL predicts emphysema progression, the systemic biological consequences detectable in peripheral blood are unknown.

Methods. We profiled $\sim 4,979$ plasma proteins using the SomaScan platform in $n = 1,306$ COPDGene participants. Differential protein expression was assessed by multivariable linear regression of each $\log_2$ -transformed protein against MAL score, adjusting for age, sex, race, smoking, pack-years, CT scanner, and ancestry principal components. Gene-set enrichment analysis (GSEA) and STRING network analysis characterised biological pathways. An adaptive LASSO was used to construct a parsimonious Proteomic Risk Score (PRS) for MAL. The PRS was then tested for association with emphysema progression (change in lung density, Phase 3 – Phase 2) using multivariable linear regression, and Akaike Information Criterion (AIC) comparisons were performed for clinical, PRS-only, and combined models. Causal mediation

analysis (medflex, natural effects framework) evaluated whether the PRS mediates the pathway $\text{MAL} \rightarrow \text{PRS} \rightarrow \text{emphysema}$ progression.

Results. Forty-five proteins were differentially expressed with MAL at $\text{FDR} < 0.05$, led by Adiponectin (ADIPOQ, $\beta = 0.27$) and Troponin T (TNNT2, $\beta = 0.19$). GSEA identified nine significant pathways, predominantly reflecting metabolic suppression (fatty acid metabolism, oxidative phosphorylation, glycolysis) and immune/inflammatory activation (chemokine receptor signalling, antimicrobial peptides). An adaptive LASSO selected 378 proteins that explained 15.7% of MAL variance in the test set. The resulting PRS was not significantly associated with emphysema progression ($\beta = 0.42$, 95% CI $[-0.05, 0.88]$, $p = 0.079$) and did not improve model fit over clinical variables alone ($\Delta\text{AIC} = +0.8$). Causal mediation showed that the PRS mediates -0.6% of the MAL effect on progression (Natural Indirect Effect = -0.12 , $p = 0.97$), while the direct effect of MAL was large and significant (Natural Direct Effect = 19.7 , $p = 0.004$).

Conclusions. Plasma proteomics reflects the metabolic and inflammatory burden of mechanically stressed lung. However, the proteomic signature of MAL does not sit causally between MAL and emphysema progression, nor does it meaningfully enrich prediction beyond standard clinical variables, suggesting that the mechanistic pathway from mechanical stress to disease progression operates independently of the circulating proteome.

1 Introduction

Chronic obstructive pulmonary disease (COPD) is characterised by progressive and irreversible emphysema driven in part by destructive mechanical forces acting on lung parenchyma. Mechanically Affected Lung (MAL) is a continuous, CT-derived score (range 0.01–0.40) that quantifies the fraction of lung volume under destructive mechanical stress ?. Higher MAL predicts emphysema progression independently of baseline lung density and standard spirometric measures, suggesting it captures a distinct pathophysiological process.

The systemic biological correlates of MAL that are detectable in peripheral plasma are unknown. Identifying plasma biomarkers of mechanical lung stress would serve two complementary goals: (1) understand the biological pathways by which mechanical forces damage the lung, and (2) develop a blood-based test to enrich patient populations for MAL and emphysema progression, enabling mechanistic studies and clinical trials.

The COPDGene cohort ? has performed SomaScan high-throughput plasma proteomics (~4,979 proteins) across more than 5,000 participants. We used this resource to: (i) map the circulating protein correlates of MAL; (ii) characterise the implicated biological pathways through gene-set enrichment and protein interaction network analyses; (iii) build a parsimonious Proteomic Risk Score (PRS) for MAL using adaptive LASSO penalised regression; and (iv) test whether this PRS is associated with emphysema progression and whether it mediates the causal effect of MAL on progression.

2 Methods

2.1 Study Population

Participants were drawn from the COPDGene study, a multi-centre longitudinal cohort. Plasma proteomic data (SomaScan), phenotype and covariate data, and CT-derived MAL scores were linked by subject identifier. The analysis cohort comprised the complete cases

with non-missing values for all analysis variables ($n = 1,306$ after three-way merge and complete-case filtering; see Table 2).

Missing variables.

- **BMI:** No BMI column was identified in the phenotype file for this release (`grep("[Bb] [Mm] [Ii]")` returned no match). BMI was therefore omitted from Table 1, the covariate set for differential expression, and all regression models. When BMI data become available, it should be added as an unpenalised covariate.
- **FEV₁% predicted (FEV1pp):** No column matching FEV1pp.P2 or related patterns (FEV1pct, FEV1_percent_pred) was found in the phenotype file. FEV₁% predicted is therefore absent from Table 1. The column FEV1_FVC_post.P2 (FEV₁/FVC ratio) was present and is included.

2.2 Proteomic Data

SomaScan relative fluorescence unit (RFU) values for $\sim 4,979$ proteins were $\log_2$ -transformed prior to all analyses:

$$\tilde{x}_{ij} = \log_2(\max(x_{ij}, 10^{-6})).$$

Protein identifiers follow the pattern `log2_X{seqID}` and were mapped to HGNC gene symbols and UniProt targets via the SomaScan annotation file (`metadat5k.csv`).

2.3 Outcomes

- **MAL score (meanmal):** continuous, range 0.01–0.40.
- **Emphysema progression:** change in CT lung density $\Delta\rho = \rho_{P3} - \rho_{P2}$ (Hounsfield units, HU), stored as `Change_lung_density_vnb_P2_P3`. Negative values indicate progression (density loss).

2.4 Differential Protein Expression

For each of the 4,979 proteins we fitted:

$$\tilde{x}_{\text{protein}} \sim \beta_0 + \beta_{\text{MAL}} \cdot \text{meanmal} + \boldsymbol{\gamma}^\top \mathbf{z} + \varepsilon,$$

where $\mathbf{z}$ comprises age, sex, race, pack-years, current smoking status, CT scanner model, and ancestry principal components (PC1–PC5). The MAL coefficient $\hat{\beta}_{\text{MAL}}$ and its t -statistic were extracted for each protein. Multiple testing was controlled by the Benjamini–Hochberg false discovery rate (FDR); proteins with $\text{FDR} < 0.05$ were considered differentially expressed.

2.5 Gene-Set Enrichment Analysis

All 4,979 proteins were ranked by their MAL t -statistic and submitted to `fgsea` using gene sets from `msigdb`: Hallmark (H), KEGG Legacy (C2:CP:KEGG), and Reactome (C2:CP:REACTOME) collections. Minimum and maximum gene-set sizes were 15 and 500, respectively. Pathways with Benjamini–Hochberg-adjusted $p < 0.05$ were reported.

2.6 STRING Network Analysis

FDR-significant proteins were mapped to the STRING v11.5 protein interaction database (STRINGdb R package, human, score threshold = 400) and the interaction network was plotted.

2.7 Proteomic Risk Score for MAL

Adaptive LASSO. We randomly split the cohort 70:30 into training ($n = 914$) and test ($n = 392$) sets (`set.seed(2024)`). A ridge regression pre-step was applied to the residualised MAL outcome (after projecting out the unpenalised covariates) to derive protein-specific

105 adaptive weights:

$$w_j = \frac{1}{|\hat{\beta}_j^{\text{ridge}}|}.$$

106 An L1-penalised regression (glmnet, $\alpha = 1$) with `penalty.factor =  $w_j$` was then fitted using
107 10-fold cross-validation; $\lambda_{\min}$ was selected. Unpenalised covariates (age, sex, race, smoking,
108 pack-years, scanner, PC1–PC5) were assigned `penalty.factor = 0`.

109 **Post-selection OLS.** Proteins with non-zero LASSO coefficients were refitted by ordinary
110 least squares (OLS) to obtain unbiased estimates, standard errors, 95% confidence intervals,
111 and p -values.

112 **Proteomic Risk Score computation.** For each subject i :

$$\text{PRS}_i = \sum_{j \in \mathcal{S}} \hat{\beta}_j^{\text{OLS}} \cdot \tilde{x}_{ij},$$

113 where $\mathcal{S}$ is the selected protein set. The PRS was standardised to mean 0, SD 1 across the
114 full cohort.

115 2.8 Association of PRS with Emphysema Progression

116 Three nested models were fitted by OLS:

117 1. **Clinical model:** $\Delta\rho \sim \text{age} + \text{sex} + \text{race} + \text{current smoking} + \text{pack-years} + \text{CT}$
118 `scanner` **[DATA NOTE: BMI omitted – column absent from phenotype file.**
119 **Add when available.]**

120 2. **PRS-only model:** $\Delta\rho \sim \text{PRS} + \text{baseline lung density (lung_density_vnb_P2)} + \text{CT}$
121 `scanner`

122 3. **Combined model:** $\Delta\rho \sim \text{PRS} + \text{age} + \text{sex} + \text{race} + \text{current smoking} + \text{pack-years}$
123 `+ CT scanner` **[DATA NOTE: BMI omitted.]**

Model fit was compared by AIC; a difference of -2 or more (lower) was considered a meaningful improvement. The PRS β coefficient, 95% confidence interval, and p -value were reported from Model 3.

2.9 Causal Mediation Analysis

The causal pathway $\text{MAL} \rightarrow \text{PRS} \rightarrow \text{emphysema progression}$ was evaluated using the `medflex` R package (natural effects framework) ?. An imputation model for the mediator was specified as:

$$\Delta\rho \sim \text{meanmal} \times \text{PRS} + \text{covariates}$$

using `neImpute()`, followed by `neModel()` with robust (sandwich) standard errors. The following quantities were estimated:

- **Natural Direct Effect (NDE)**: effect of MAL on progression *not* through the protein risk score
- **Natural Indirect Effect (NIE)**: effect of MAL on progression *through* the protein risk score (i.e., mediation)
- **Total effect**: NDE + NIE
- **Proportion mediated**: NIE / Total

An analogous analysis was performed for each protein in the risk score individually (Supplementary Table S2).

2.10 Software

All analyses were performed in R (version ≥ 4.0) using: `glmnet` (adaptive LASSO), `pROC` (AUC), `fgsea` and `msigdbR` (GSEA), `STRINGdb` (network), `medflex` (mediation), `gtsummary` (Table 1), `ggplot2` and `ggrepel` (figures). Full reproducible code is available in the accompanying Jupyter notebook `MAL_omics/MAL_omics_proteomics.ipynb`.

3 Results

3.1 Study Population

After three-way linkage of proteomic, phenotype, and MAL data and complete-case filtering, 1,306 participants were included. Demographic and clinical characteristics are summarised in Table 2.

Note on missing clinical variables. FEV₁% predicted and BMI were not available in the current phenotype file extract and are absent from Table 1. Neither variable was included as a covariate in any regression model. Analysts should verify whether these columns exist under alternative names in other phenotype releases and update the analysis accordingly.

3.2 Differential Protein Expression with MAL

Of 4,979 proteins tested, **45 were significantly associated with MAL** at FDR < 0.05 (Figure ??). The strongest positive associations (higher protein with higher MAL) were observed for Adiponectin (*ADIPOQ*, $\beta = 0.27$, FDR = 4.9×10^{-4}) and Cardiac Troponin T (*TNNT2*, $\beta = 0.19$, FDR = 7.8×10^{-4}). Additional proteins positively associated with MAL included ADAMTS-5 (*ADAMTS5*), NCAM-120 (*NCAM1*), PAK7, MIA, and Chormogranin B (*CHGB*).

The strongest negative associations (lower protein with higher MAL) were Proctasin (*PRSS8*, $\beta = -0.12$), Aminoacylase-1 (*ACY1*, $\beta = -0.26$), Nogo Receptor (*RTN4R*, $\beta = -0.15$), RET ($\beta = -0.18$), ADH4 ($\beta = -0.25$), and SFRP4 ($\beta = -0.12$). Full results for all 4,979 proteins are provided in Supplementary Table S1.

3.3 Biological Pathway Enrichment (GSEA)

Nine pathways reached significance (adjusted $p < 0.05$), summarised in Figure ?? and Table 1:

Table 1: Significant GSEA pathways ($p_{adj} < 0.05$). NES < 0 indicates depletion in high-MAL subjects.

Collection	Pathway	NES	p_{adj}
Hallmark	Fatty Acid Metabolism	-1.76	0.021
Hallmark	Oxidative Phosphorylation	-1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	-2.02	0.018
Reactome	Biological Oxidations	-2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	-1.84	0.013
Reactome	Diseases of Metabolism	-1.76	0.032
Reactome	Phase I Functionalization of Compounds	-1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

Seven pathways were **depleted** in individuals with high MAL, all reflecting core metabolic functions: energy metabolism (oxidative phosphorylation, glycolysis), lipid catabolism (fatty acid metabolism), amino acid handling, and xenobiotic processing (Phase I, biological oxidations). Two pathways were **enriched**: chemokine receptor signalling (NES = 2.02) and antimicrobial peptide activity (NES = 1.94), consistent with a shift toward an inflammatory state.

3.4 STRING Protein Interaction Network

FDR-significant proteins were mapped to the STRING v11.5 human protein-protein interaction network (score threshold ≥ 400). The resulting network is shown in Figure ??.

[DATA NOTE: Describe network topology and key hubs once the figure is reviewed: e.g., number of nodes, number of edges, main cluster identities.]

3.5 Proteomic Risk Score for MAL

Adaptive LASSO selected **378 proteins** from 4,979 candidates. On the held-out test set the score explained **15.7%** of MAL variance ($R^2 = 0.157$). The top 20 proteins by absolute OLS β -coefficient are listed in Table 5. Strongest positive contributors included ZBP1 ($\hat{\beta} = 0.29$), GREM2 ($\hat{\beta} = 0.19$), Complement C7 ($\hat{\beta} = 0.17$), and Collagen Type III (*COL3A1*,

$\hat{\beta} = 0.13$). Strongest negative contributors included Complement Factor I (CFI , $\hat{\beta} = -0.23$), SEC13 ($\hat{\beta} = -0.15$), and Spermine Synthase (SMS , $\hat{\beta} = -0.13$). Full weights for all 378 proteins are provided in Supplementary Table S2.

3.6 PRS Association with Emphysema Progression

Results of the three-model AIC comparison are shown in Table 3. The clinical model achieved AIC = 8,815. The PRS-only model was substantially *worse* (AIC = 8,848, $\Delta\text{AIC} = +33.1$), indicating that a proteomic-only model lacking standard clinical predictors performs poorly. Adding the PRS to the clinical model provided essentially no improvement (AIC = 8,816, $\Delta\text{AIC} = +0.8$ vs. clinical alone); this falls well short of the -2 threshold for a meaningful gain.

In the combined model, the PRS coefficient was $\hat{\beta} = 0.42$ HU per SD (95% CI $[-0.05, 0.88]$, $p = 0.079$), indicating a modest, statistically non-significant trend.

The quartile analysis (Figure ??) shows the PRS-associated effect size on emphysema progression across risk score quartiles.

3.7 Causal Mediation

Results are reported in Table 4. The Natural Direct Effect of MAL on emphysema progression (the pathway *not* through the protein risk score) was large and statistically significant: NDE = 19.7 HU ($p = 0.004$). In contrast, the Natural Indirect Effect (the mediated pathway through the PRS) was negligible: NIE = -0.12 HU ($p = 0.97$). The total effect was 19.6 HU, and the proportion mediated was -0.6% , effectively zero.

These results indicate that **the plasma proteomic signature of MAL does not mediate the effect of MAL on emphysema progression**. The dominant pathway is direct and independent of the circulating proteome.

For individual proteins in the risk score, TREM2 showed the largest proportion mediated (13.2%, NIE $p = 0.017$), followed by SEZ6L (16.8%, $p = 0.025$) and RSPO1 (13.8%,

210 $p = 0.038$); however, all individual indirect effects were small in absolute magnitude (Sup-
211 plementary Table S3).

212 4 Discussion

213 We characterised the plasma proteomic landscape of mechanically stressed lung in a large,
214 well-phenotyped cohort. Three principal findings emerge.

215 **First**, MAL is associated with a distinct plasma protein signature. The dominant
216 signals—elevated adiponectin and troponin T, suppressed metabolic enzymes (ADH4, ACY1)—
217 are consistent with a state of systemic metabolic stress and mechanical tissue damage. The
218 positive association of ADAMTS5 (a matrix metalloproteinase involved in aggrecan degra-
219 dation) and the negative association of SFRP4 (a Wnt-signalling inhibitor) suggest dysregu-
220 lated extracellular matrix remodelling. Pathway analysis reinforces this picture: high-MAL
221 individuals show broad suppression of aerobic metabolism and upregulation of innate im-
222 mune pathways.

223 **Second**, despite selecting 378 proteins, the MAL PRS explained only 15.7% of MAL
224 variance, and did not significantly predict emphysema progression ($p = 0.079$) or improve
225 AIC beyond clinical variables alone ($\Delta\text{AIC} = +0.8$). This suggests that the plasma proteome
226 reflects, but does not drive, the mechanical lung phenotype at a level detectable above clinical
227 noise.

228 **Third**, causal mediation analysis showed that the proteomic signature mediates essen-
229 tially none (-0.6%) of the pathway from MAL to emphysema progression. The direct effect
230 is overwhelmingly dominant ($\text{NDE} = 19.7$, $p = 0.004$). This implies that mechanical stress
231 causes progression through mechanisms that are not captured by the peripheral proteome—
232 perhaps local tissue processes (alveolar wall stress, local inflammation) that do not produce
233 detectable systemic signals at this level of assay sensitivity.

Limitations. BMI and FEV₁% predicted were absent from the current phenotype file extract and could not be included as covariates. Their absence may have introduced residual confounding in differential expression and regression models. Additionally, SomaScan measures circulating aptamer-bound proteins; it does not capture local lung tissue proteomics, which may be more directly relevant to the mechanism. The $R^2 = 0.157$ for MAL prediction indicates considerable unexplained variance, likely reflecting the limitation of blood proteomics as a proxy for a CT-derived structural measure.

Tables

Table 2: Demographic and clinical characteristics of the study population ($n = 1,306$). Data are mean (SD) for continuous variables and n (%) for categorical variables. Emphysema progression is Phase 3 – Phase 2 CT lung density (HU); negative values indicate progression.

Characteristic	Overall ($n = 1,306$)
<i>Demographics</i>	
Age, years	[TODO: mean (SD)]
Sex, female	[TODO: n (%)]
Race, White	[TODO: n (%)]
<i>Smoking</i>	
Current smoker	[TODO: n (%)]
Pack-years	[TODO: mean (SD)]
<i>Pulmonary function</i>	
FEV ₁ % predicted	[DATA NOTE: Column FEV1pp_P2 not found in phenotype file; omitted]
FEV ₁ /FVC	[TODO: mean (SD)]
<i>CT imaging</i>	
Lung density Phase 2 (HU)	[TODO: mean (SD)]
MAL score	[TODO: mean (SD)]
Emphysema progression (HU)	[TODO: mean (SD)]
<i>Anthropometric</i>	
BMI (kg/m ²)	[DATA NOTE: BMI column not found in phenotype file; omitted]

Note. Values are populated from the `tbl_summary()` output in the executed notebook.

Abbreviations: MAL, Mechanically Affected Lung; FEV₁, forced expiratory volume in 1 s;

FVC, forced vital capacity; HU, Hounsfield units.

Table 3: AIC comparison of regression models for emphysema progression. Δ AIC is relative to the clinical model. An improvement of ≤ -2 is considered meaningful.

Model	AIC	Δ AIC
Clinical (age, sex, race, smoking, pack-years, scanner) ^a	8,815	0
PRS only (PRS + baseline density + scanner)	8,848	+33.1
Clinical + PRS	8,816	+0.8

^aBMI omitted (column not available in current phenotype extract).

PRS $\hat{\beta} = 0.42$ HU/SD, 95% CI $[-0.05, 0.88]$, $p = 0.079$ in combined model.

Table 4: Causal mediation analysis: MAL $\rightarrow$ Proteomic Risk Score $\rightarrow$ Emphysema Progression. Estimates from the natural effects model (medflex) with robust (sandwich) standard errors.

Parameter	Estimate (HU)	z	p -value
Natural Direct Effect (NDE)	19.72	2.92	0.004
Natural Indirect Effect (NIE)	-0.12	-0.04	0.971
Total effect	19.60	—	—
Proportion mediated		-0.6%	

Covariates: age, sex, race, pack-years, current smoking, CT scanner, baseline lung density. BMI omitted (not available). Results should be revisited when BMI data are added.

Table 5: Top 20 proteins in the MAL Proteomic Risk Score by absolute OLS $\hat{\beta}$. CI, 95% confidence interval.

Rank	Gene	Target	$\hat{\beta}$	95% CI	p
1	ZPBP	ZPBP1	+0.286	[0.160, 0.412]	< 0.001
2	CFI	Factor I	-0.226	[-0.398, -0.054]	0.010
3	GREM2	GREM2	+0.193	[0.113, 0.273]	< 0.001
4	C7	Complement C7	+0.169	[0.030, 0.309]	0.017
5	SEC13	SEC13	-0.153	[-0.254, -0.053]	0.003
6	SERPINF2	α_2 -Antiplasmin	+0.149	[0.011, 0.287]	0.035
7	FSTL1	FSTL1	-0.147	[-0.309, 0.015]	0.075
8	APOH	β_2 -Glycoprotein I	+0.143	[-0.049, 0.335]	0.145
9	COL3A1	Collagen Type III	+0.132	[0.061, 0.204]	< 0.001
10	C3	C3a	+0.131	[0.021, 0.242]	0.020
11	SMS	Spermine Synthase	-0.125	[-0.179, -0.072]	< 0.001
12	MATN2	MATN2	-0.125	[-0.242, -0.008]	0.036
13	GALNT11	GLT11	-0.122	[-0.225, -0.019]	0.021
14	ROBO2	ROBO2	-0.119	[-0.220, -0.017]	0.022
15	SEZ6L	SEZ6L	+0.117	[0.020, 0.214]	0.019
16	FBLN1	Fibulin-1	-0.113	[-0.278, 0.051]	0.175
17	ADAMTS5	ADAMTS-5	+0.113	[0.062, 0.163]	< 0.001
18	CCL23	MPIF-1	-0.112	[-0.160, -0.063]	< 0.001
19	CREB3L4	CR3L4	+0.108	[0.047, 0.169]	< 0.001
20	CNTN1	Contactin-1	-0.107	[-0.229, 0.015]	0.086

Figure Legends

Figure 1 — Study Design Schematic. Overview of the analysis workflow. COPDGene participants with SomaScan plasma proteomics, CT-derived MAL scores, and longitudinal lung density measurements were studied. Differential expression identified proteins associated with MAL; pathway and network analyses characterised their biology; adaptive LASSO constructed a Proteomic Risk Score (PRS); and the PRS was tested for association with and mediation of emphysema progression. **[DATA NOTE: Figure 1 is a conceptual schematic; to be drawn as a diagram (e.g., BioRender or vector graphics).]**

Figure 2 — Protein Expression Landscape of MAL. (A) Volcano plot. Each point is one protein ($n = 4,979$). The x -axis shows the MAL $\hat{\beta}$ coefficient; the y -axis shows

$-\log_{10}(\text{FDR})$. Red: $\text{FDR} < 0.05$ and $\hat{\beta} > 0$ (enriched with MAL); blue: $\text{FDR} < 0.05$ and $\hat{\beta} < 0$ (depleted with MAL); grey: not significant. Top 20 proteins by FDR are labelled. **(B) GSEA barplot.** Horizontal bars show the normalised enrichment score (NES) for all 9 significant pathways ($\text{padj} < 0.05$), coloured by direction and faceted by gene-set collection. **(C) STRING network.** Protein–protein interaction network for the 45 FDR-significant proteins (STRING v11.5, score ≥ 400). Node colour reflects the sign of the MAL association.

Figure 3 — Quartile Plot of Proteomic Risk Score. Effect size (OLS $\hat{\beta}$ for the PRS predictor, HU/SD) with 95% confidence intervals from separate regression models of emphysema progression on the PRS within each risk-score quartile. The x -axis shows PRS quartile (1 = lowest, 4 = highest); the y -axis shows the $\hat{\beta}$ coefficient. The dashed horizontal line marks the null.

Supplement

Supplementary Tables

Table S1. Complete differential expression results for all 4,979 proteins (gene, target, $\hat{\beta}$, SE, t -statistic, p -value, FDR). File: `MAL_omics/mal_de_results.csv`.

Table S2. Proteomic Risk Score weights for all 378 selected proteins ($\hat{\beta}_{\text{OLS}}$, 95% CI, p -value). File: `MAL_omics/mal_risk_score_weights.csv`.

Table S3. Per-protein mediation results for the 378 PRS proteins (NDE, NIE, total effect, proportion mediated, NIE p -value). File: `MAL_omics/mal_mediation_supp.csv`.

Supplementary Figures

Supplementary Figure A. Forest plot of the top 30 FDR-significant proteins and their $\hat{\beta}$ coefficients for MAL association (HU per $\log_2$ RFU unit), with 95% confidence intervals.

Proteins are ordered by effect size.

Supplementary Figure B. Forest plot of the same 30 FDR-significant proteins and their $\hat{\beta}$ coefficients for emphysema progression association (HU per $\log_2$ RFU unit, adjusted for baseline lung density and covariates), with 95% confidence intervals.

Assumptions and Data Quality Notes

The following issues were encountered during analysis and should be addressed before finalising the manuscript:

1. **BMI not found.** No column matching BMI was detected in `pheno_MAL.csv`. The COPDGene data dictionary should be consulted for the correct column name (e.g., `BMI_P2`, `bmi_P2`). Once found, add `penalty.factor = 0` for BMI in the adaptive LASSO and include it in the clinical model and Table 1.
2. **FEV₁% predicted not found.** The expected column `FEV1pp_P2` does not exist. Possible alternatives: `FEV1pct_pred_P2`, `FEV1_PP_P2`. FEV₁/FVC (`FEV1_FVC_post_P2`) was found and used where spirometry was required.
3. **Progressor definition.** Emphysema progression is defined continuously as $\Delta\rho$ (HU). A binary definition (e.g., $\Delta\rho < -1.5$ HU) could be used for sensitivity analyses and would enable AUC/ROC-based model comparison.
4. **R² for MAL PRS.** $R^2 = 0.157$ on the test set is modest. This may reflect the inherent limitations of blood proteomics for a CT-derived structural phenotype, or insufficient power in the current split. A bootstrap-stabilised estimate should be reported alongside the single test-set value.
5. **Figure 1 schematic.** A study design figure has not been automatically generated; it requires manual preparation (e.g., BioRender).